KB금융그룹 | 국민의 평생 금융파트너

K-디지털 트레이닝

vue-router를 이용한 라우팅 1

[KB] IT's Your Life

KB국민은행

SPA(단일 페이지 애플리케이션: Single Page Application)

- 하나의 페이지 안에서 데스크톱 애플리케이션과 같은 사용자 경험을 제공
- 여러 화면을 하나의 페이지 안에서 제공, 화면을 별도로 로딩하지 않음

→ 화면마다 고유의 식별자(URI)를 기반으로 화면을 렌더링해야 함

- 요청한 URI 경로에 따라 각각 다른 화면이 렌더링되도록 함

→ vue-router 라이브러리 이용

createRouter()

- router 객체를 생성
  - URI와 이 경로를 처리할 컴포넌트 매핑

```
import { createRouter, createWebHistory } from 'vue-router'
...
const router = createRouter({
    history: createWebHistory(),
    routes: [
        { path: '/', component: Home },
        { path: '/about', component: About },
        { path: '/members', component: Members },
        { path: '/videos', component: Videos},
    ]
});
```

router 객체의 등록

```
const app = createApp(App)
app.use(router)
app.mount('#app')
```

<RouterView>

- 각 경로별 컴포넌트를 렌더링할 위치를 지정하는 컴포넌트

```
<template>
  <div class="container">
    <Header />
    <RouterView/>
  </div>
</template>
```

<RouterLink to="경로">

- 화면 전환 을 위한 링크 생성

<router-link to="[등록시킬 URI 경로]">[링크 텍스트]</router-link>

```
<RouterLink to="/">Home</RouterLink>
```

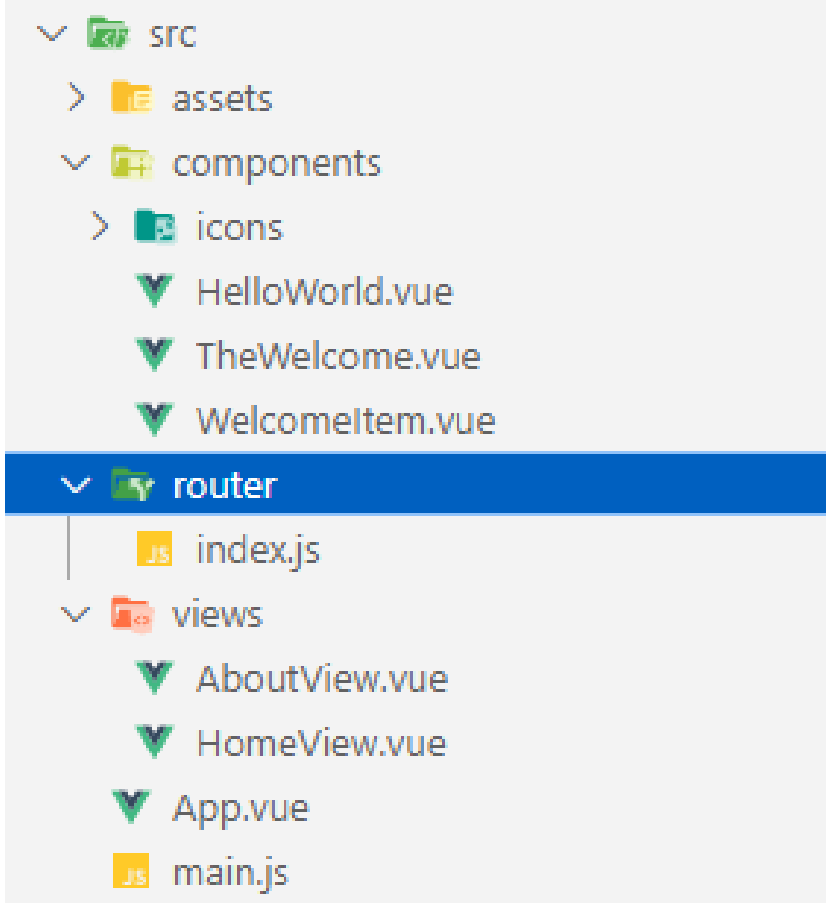
실습 프로젝트 생성

```
npm init vue router-test

┌ Vue.js - The Progressive JavaScript Framework
│
◆ Select features to include in your project: (↑/↓ to navigate, space to select, a to toggle all, enter to confirm)
│  ◻ TypeScript
│  ◻ JSX Support
│  ◼ Router (SPA development)
│  ◻ Pinia (state management)
│  ◻ Vitest (unit testing)
│  ◻ End-to-End Testing
│  ◻ ESLint (error prevention)
│  ◻ Prettier (code formatting)
└...
cd router-test
npm install
```

package.json

```
{
  "name": "router-test",
  "version": "0.0.0",
  "private": true,
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "preview": "vite preview"
  },
  "dependencies": {
    "vue": "^3.3.11",
    "vue-router": "^4.2.5"
  },
  "devDependencies": {
    "@vitejs/plugin-vue": "^4.5.2",
    "vite": "^5.0.10"
  }
}
```

기본 골격


src
assets
components
icons
HelloWorld.vue
TheWelcome.vue
WelcomeItem.vue
router
index.js
views
AboutView.vue
HomeView.vue
App.vue
main.js

src/router/index.js

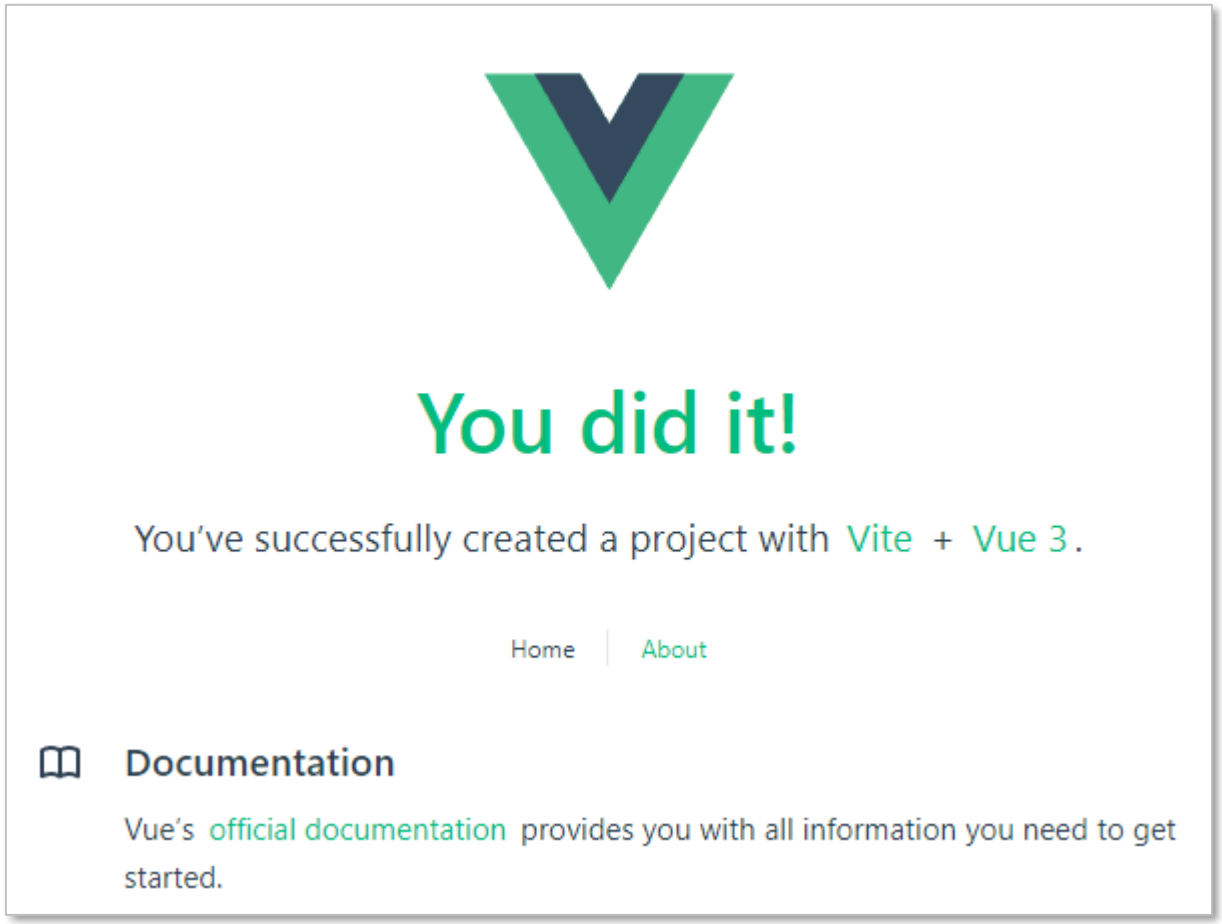
```
import { createRouter, createWebHistory } from 'vue-router'
import HomeView from '../views/HomeView.vue'

const router = createRouter({
  history: createWebHistory(import.meta.env.BASE_URL),
  routes: [
    {
      path: '/',
      name: 'home',
      component: HomeView
    },
    {
      path: '/about',
      name: 'about',
      // route level code-splitting
      // this generates a separate chunk (About.[hash].js) for this route
      // which is lazy-loaded when the route is visited.
      component: () => import('../views/AboutView.vue')
    }
  ]
})
export default router
```

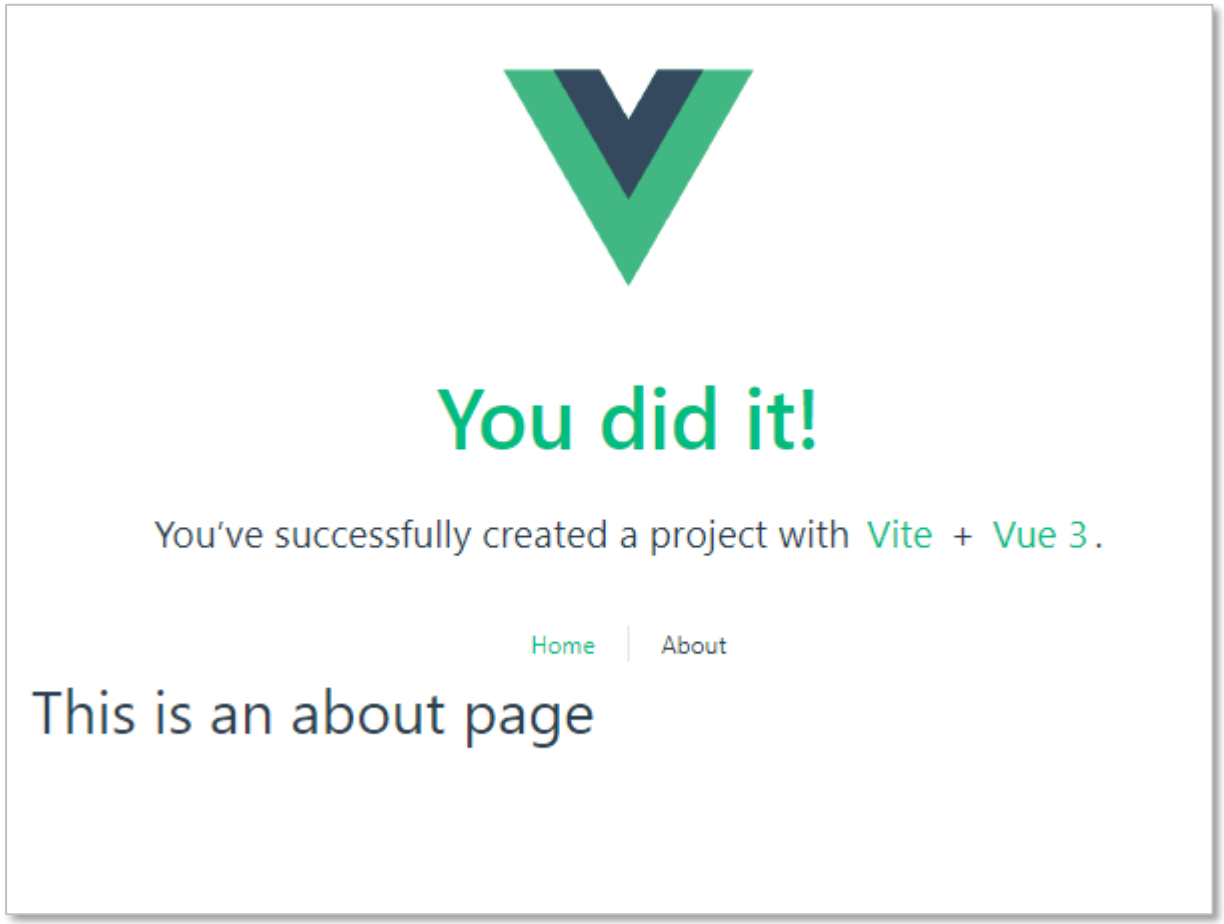
src/main.js

```
import './assets/main.css'

import { createApp } from 'vue'
import App from './App.vue'
import router from './router'

const app = createApp(App)

app.use(router)

app.mount('#app')
```

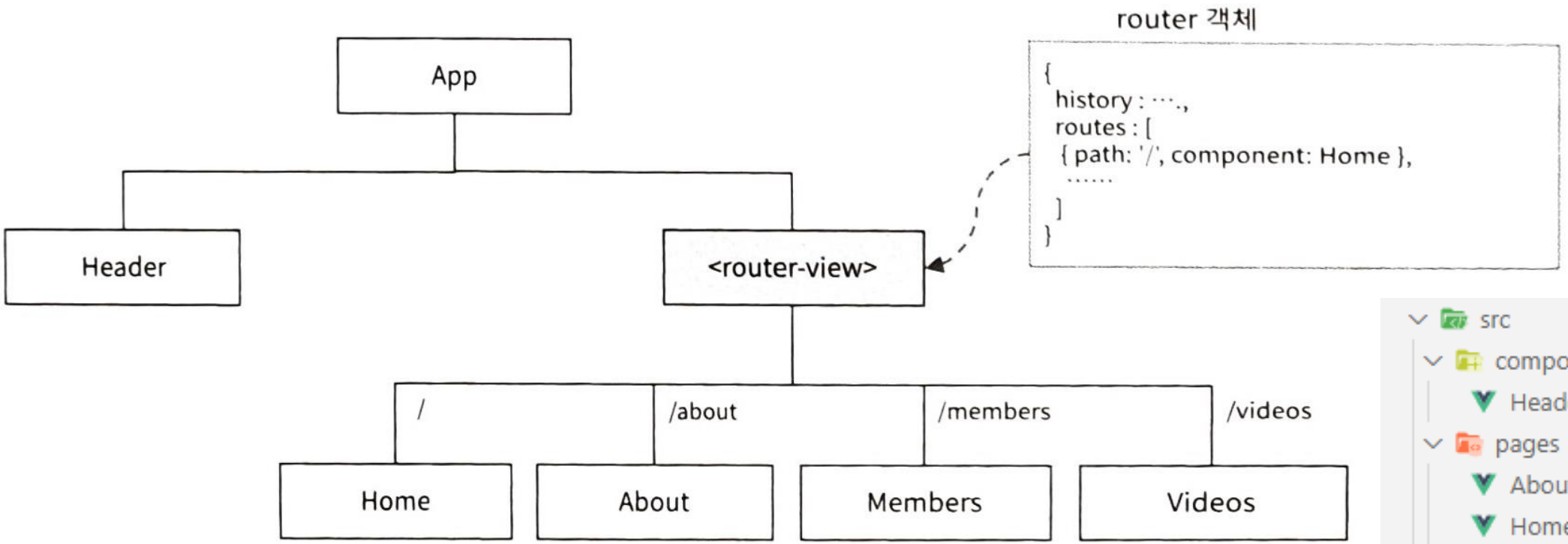
src/App.vue

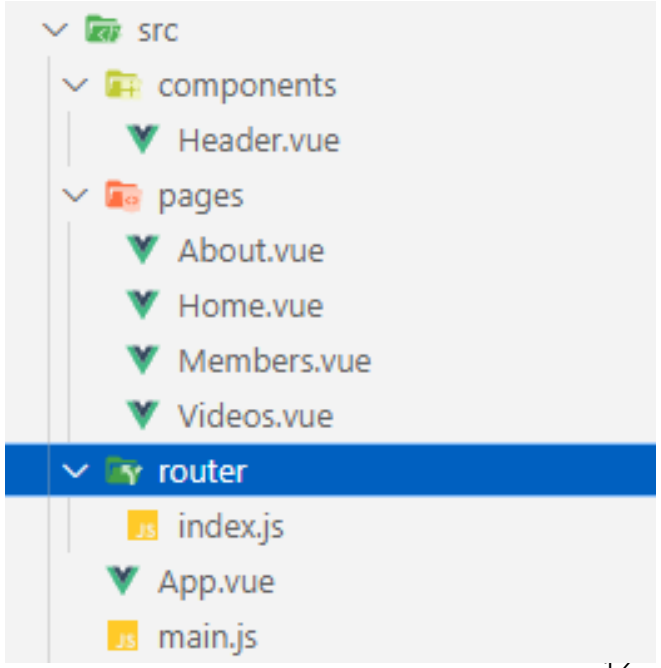
```
<script setup>
import { RouterLink, RouterView } from 'vue-router'
import HelloWorld from './components/HelloWorld.vue'
</script>

<template>
  <header>
    <img alt="Vue logo" class="logo" src="@/assets/logo.svg" width="125" height="125" />

    <div class="wrapper">
      <HelloWorld msg="You did it!" />

      <nav>
        <RouterLink to="/">Home</RouterLink>
        <RouterLink to="/about">About</RouterLink>
      </nav>
    </div>
  </header>

  <RouterView />
</template>
<style scoped>
…
</style>
```


You did it!
You've successfully created a project with Vite + Vue 3.
Home
About
Documentation
Vue's official documentation provides you with all information you need to get started.


You did it!
You've successfully created a project with Vite + Vue 3.
Home
About
This is an about page

작업할 프로젝트 컴포넌트 구조





부트스트랩5 설치

```
npm install bootstrap@5
```

page 컴포넌트 정의

- src/pages
  - Home.vue
  - About.vue
  - Members.vue
  - Videos.vue

  - name과 타이틀을 각 페이지마다 지정

```
<template>
    <div class="card card-body">
        <h2>Home</h2>
    </div>
</template>

<script>
export default {
  name : "Home"
}
</script>
```

src/router/index.js

```
import { createRouter, createWebHistory } from 'vue-router'

import Home from '@/pages/Home.vue'
import About from '@/pages/About.vue'
import Members from '@/pages/Members.vue'
import Videos from '@/pages/Videos.vue'

const router = createRouter({
    history: createWebHistory(),
    routes : [
        { path: '/', component: Home },
        { path: '/about', component: About },
        { path: '/members', component: Members },
        { path: '/videos', component: Videos },
    ]
})

export default router;
```

src/main.js

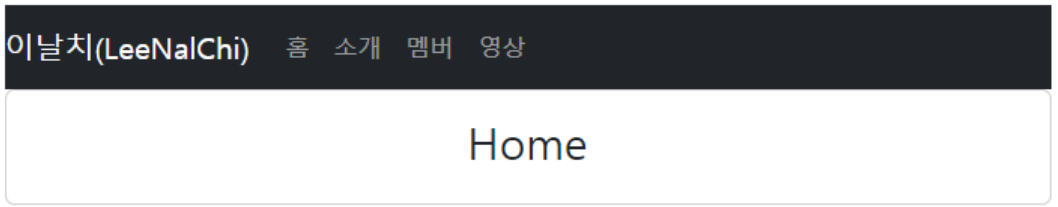
```
import { createApp } from 'vue'
import 'bootstrap/dist/css/bootstrap.css'
import App from './App.vue'
import router from './router'

const app = createApp(App)
app.use(router)
app.mount('#app')
```

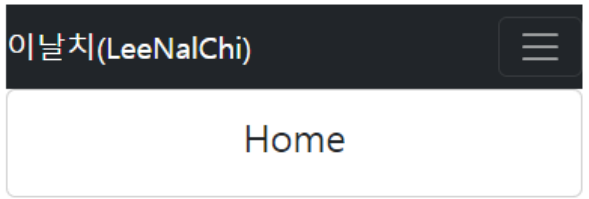
src/components/Header.vue

```
<template>
  <nav class="navbar navbar-expand-md bg-dark navbar-dark mt-2">
    <span class="navbar-brand">이날치(LeeNalChi)</span>
    <button class="navbar-toggler" type="button" @click="changeIsNavShow">
      <span class="navbar-toggler-icon"></span>
    </button>

    <div :class="navClass">
      <ul class="navbar-nav">
        <li class="nav-item">
          <router-link class="nav-link" to="/">홈</router-link>
        </li>
        <li class="nav-item">
          <router-link class="nav-link" to="/about">소개</router-link>
        </li>
        <li class="nav-item">
          <router-link class="nav-link" to="/members">멤버</router-link>
        </li>
        <li class="nav-item">
          <router-link class="nav-link" to="/videos">영상</router-link>
        </li>
      </ul>
    </div>
  </nav>
</template>
```

src/components/Header.vue

```
<script>
import { reactive, computed } from 'vue';

export default {
  setup() {
    const state = reactive({ isNavShow: false });

    const navClass = computed(() =>
      state.isNavShow
        ? 'collapse navbar-collapse show'
        : 'collapse navbar-collapse'
    );

    const changeIsNavShow = () => { state.isNavShow = !state.isNavShow; };

    return { state, changeIsNavShow, navClass };
  },
};
</script>
```

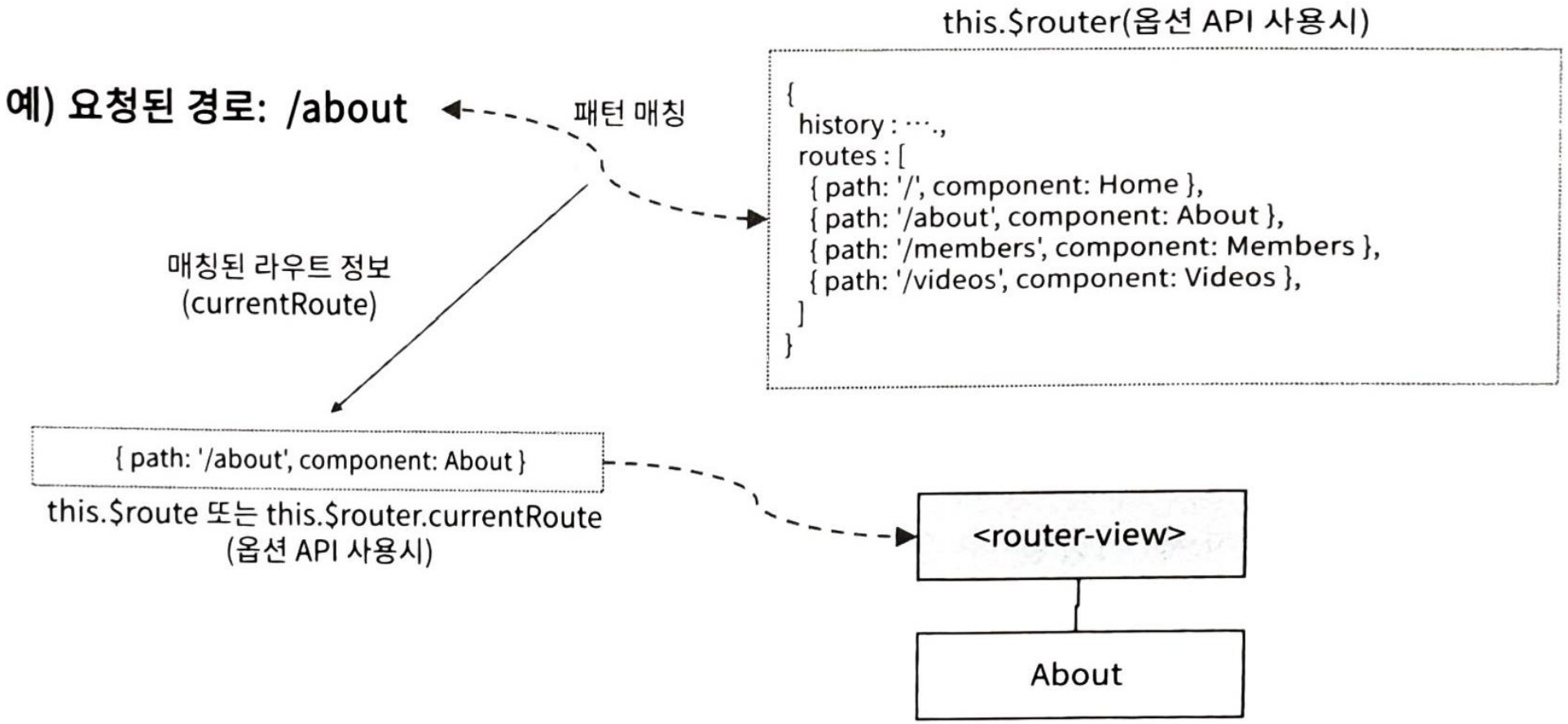




src/App.vue

```
<template>
  <div class="container">
    <Header />
    <router-view></router-view>
  </div>
</template>

<script>
import Header from '@/components/Header.vue'

export default {
  name : "App",
  components : { Header },
}
</script>
<style>
.container {
  text-align: center;
  margin-top:10px;
}
</style>
```

<router-view>의 컴포넌트 마운트 방법



vue-router 기본 적용 방법 예제 실행

이날치(LeeNalChi) 홈 소개 멤버 영상

About

Elements Vue Console Sources Network Performance Memory

App Components Timeline

Find components... <About> Filter state...

```
<App>
  <Header>
    <RouterLink>
    <RouterLink>
    <RouterLink>
    <RouterLink>
  <RouterView>
    <About>
```

```
provided
  Symbol(router view depth): undefined (Computed)
  Symbol(router view location matched): Object (Compute
  Symbol(router view location): Object (Computed)
    fullPath: "/about"
    path: "/about"
    query: Object (empty)
    hash: ""
    name: undefined
    params: Object (empty)
    matched: Array[1]
    meta: Object (empty)
    redirectedFrom: undefined
    href: "/about"
```

Elements Vue Console Sources Network Performance Memory

App Components Timeline

Find components... <RouterView> Filter state...

```
<App>
  <Header>
    <RouterLink>
    <RouterLink>
    <RouterLink>
    <RouterLink>
  <RouterView>
    <About>
```

```
props
  name: "default"
  route: undefined
provided
  Symbol(router view depth): undefined (Computed)
  Symbol(router view location matched): Object (Compute
  Symbol(router view location): Object (Computed)
    fullPath: "/about"
    path: "/about"
    query: Object (empty)
    hash: ""
    name: undefined
    params: Object (empty)
    matched: Array[1]
    meta: Object (empty)
    redirectedFrom: undefined
    href: "/about"
```

컴포넌트에서 router 객체 접근하기

- this.$router
  - src/router/index.js에서 작성한 router 객체
  - 전체 라우트 정보 파악
  - push(), replace(), go() 등의 메서드 이용가능

- Router 관련 정보 접근

구분	Options API	Composition API
라우터 객체	this.$router	const router = useRouter()
매칭된 라우트 (currentRoute)	this.$route – 또는 – this.$router.currentRoute	const currentRoute = useRoute()

컴포넌트에서 router 객체 접근하기

currentRoute 객체의 주요 속성

구분	설명
fullPath	전체 요청 경로, 쿼리문자열까지 포함(ex: /about?a=1&b=2)
matched	vue-router 객체의 routes 배열의 라우트 중 매칭된 라우트
params	URI 경로에 동적으로 전달된 파라미터 정보, 자세한 내용은 다음 절(동적 라우트)에서 다룸
path	요청 URI 경로
query	쿼리 문자열 정보 ?a=1&b=2로 요청되었다면 this.$route.query는 { a:1, b:2 }와 같은 객체임
redirectedFrom	다른 경로에서 리디렉트된 경우 리디렉트시킨 URI 경로 정보를 포함

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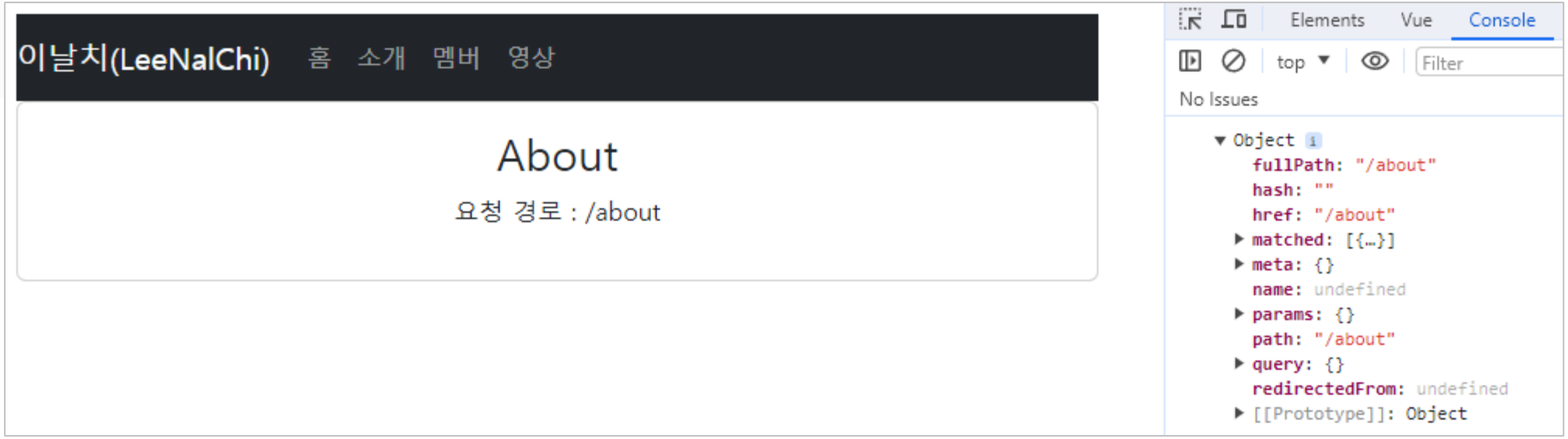
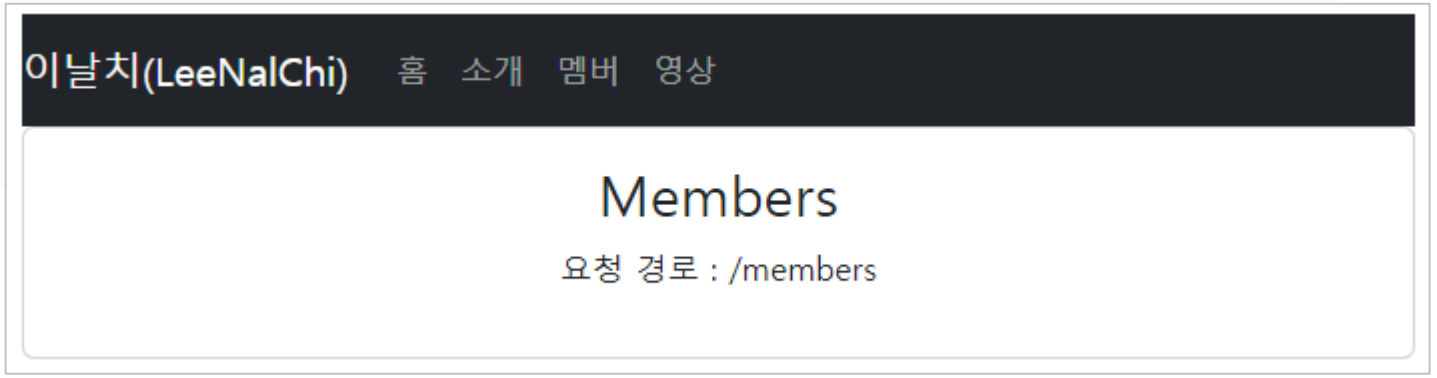
src/pages/About.vue

```
<template>
    <div class="card card-body">
        <h2>About</h2>
        <p>요청 경로 : {{$route.fullPath}}</p>
    </div>
</template>

<script>
export default {
  name : "About",
  created() {
    console.log(this.$route)
  }
}
</script>
```



src/pages/Members.vue

```
<template>
    <div class="card card-body">
        <h2>Members</h2>
        <p>요청 경로 : {{currentRoute.fullPath}}</p>
    </div>
</template>

<script>
import { useRoute } from 'vue-router'

export default {
  name : "Members",
  setup() {
    const currentRoute = useRoute();
    return { currentRoute };
  }
}
</script>
```